Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Draft for review. Technical content is filled. Resolve every CONFIRM item and complete the leader declaration before submitting.

Field	Response
Team Name	Chill Chill
Selected Domain	Domain 1 (selected by the team).
Selected SDG Track (1-17)	Proposed primary: SDG 4 - Quality Education, Target 4.4. Secondary alignment: SDG 8, Target 8.5. CONFIRM registered track.
Project / Solution Name	KerjayaKu - AI Career Exploration and Learning Companion
Name (Leader)	Edixon Teo Zan Wei
Email & Phone (Leader)	edixontzw@gmail.com | 010 880 9196
University Affiliation(s)	Taylor's University
Number of Team Members	5

Submission deadline: 25 October 2026, as shown on the supplied proposal template. Draft prepared: 25 October 2026.

2. Executive Summary & Problem Statement

2.1 The Problem Context

A student may know which job they want but still struggle to work out what to learn next. Job advertisements list requirements, course platforms offer long catalogues, and career advice sits elsewhere. Connecting these pieces takes time. Learners must judge their own skills, identify gaps and stay motivated without much individual guidance.

In Malaysia, DOSM reported that 32.2% of graduates experienced skill-related underemployment in 2024 [R1]. This national figure helps explain why career preparation matters. It does not measure our users, and we cannot claim that our prototype will reduce it.

KerjayaKu addresses a smaller, practical part of this problem: turning a career goal or job advertisement into a learning plan that someone can actually follow. We want learners to understand why a skill matters, know where to start and see the effort they have put in. Students with limited access to personal career guidance are a priority.

The intended benefit is more informed learning and better preparation for skilled work. KerjayaKu supports independent learning; it does not replace a teacher, certify a learner's competence or guarantee a job. 179 words / maximum 250

2.2 The Proposed AI Solution

KerjayaKu turns career preparation into a 3D world that learners can explore. A user describes their skills or uploads a CV, then chooses a career or adds a job advertisement. The application compares their profile with the target requirements and proposes a staged learning roadmap. Virtual mentors answer questions, and a career pet shows progress towards the relevant skills. Users can switch between English, Chinese and Bahasa Melayu.

AI handles the parts that need interpretation: extracting skills, reading job requirements, suggesting learning steps and making sense of learning reports. These calls run on the server through DeepSeek; the default model identifier in the source is deepseek-flash. The application checks structured outputs before its own rules update progress. Through Telegram, the demonstration player can report learning without opening the game. They can also ask questions using publicly sourced expert perspectives, clearly identified as AI interpretations rather than the real people.

The architecture connects a Three.js/Vite frontend to a Node.js API and a Python messaging service. JSON and SQLite support development, with optional AWS persistence. A separate collection service imports sampled Malaysian advertisements from LinkedIn and Ricebowl, using public-page/browser access or configured JSearch access. The updated market interface includes historical context, FDE and refresh controls. RunPod is an optional AI provider; its model and deployment still need confirmation.

Our Phase 2 demonstration will follow a learner from profile setup to an updated pet after a learning report. All five members completed the required GenAI, Database and Cloud Deployment workshops. The goal is to help people plan and sustain independent learning, without promising teaching or employment outcomes.

264 words / maximum 300

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

The browser game uses JavaScript, HTML and Stylus/CSS, with Three.js for rendering, Rapier for physics, GSAP for animation and Howler for audio. Vite handles development and builds. A Node.js HTTP API manages game state and makes server-side AI calls.

A separate Python service, built with FastAPI/Uvicorn and HTTP clients, connects Telegram to the game. n8n workflow definitions cover message forwarding, daily summaries and sampled Malaysian job searches through JSearch/RapidAPI. Blender/Python and glTF tools support character and pet assets.

In development, JSON files hold game state and market snapshots. SQLite stores chatbot records and the collection service's observations. The Node.js 24+ collector supports JSearch and public/browser sources through Playwright. Python data-processing scripts use pandas and Beautiful Soup. The optional AWS design uses API Gateway, Lambda and Aurora PostgreSQL through the RDS Data API, supported by Secrets Manager, S3 and CloudWatch. Its deployment has not been verified for this submission. RunPod support is implemented through an OpenAI-compatible endpoint; the deployed model and account are not established. Replit is the team's hosting platform; the chatbot runs separately from the game/API. Appendix A shows the connections.

CONFIRM whether any API access, cloud account, credits or mentoring came from an official SDC sponsor. The presence of a service in the code does not establish who supplied it.

3.2 Design-Thinking & Deployment Strategy

We started with a design hypothesis: a learner needs a clear next step and a way to see progress. The prototype follows that idea from skill profiling through target selection, gap review and learning reports. Mentors and the pet keep these steps connected within the game. Three languages make the interface more approachable.

For Phase 2, we propose a pilot with 10 consenting student volunteers using synthetic profiles. We will observe where they get stuck, compare planning time with a manual workflow and revise the wording, navigation and roadmap. This study is planned; we are not presenting interviews or usability results as completed work.

For deployment, build the game/API and serve it over HTTPS, run the chatbot separately and configure the integrations needed for the demonstration. Check saved-state recovery, AI failures, duplicate messages and Telegram authorization. Keep a clearly labelled demo mode for service outages. Run the backend/workflow tests and production build, then verify gameplay in a browser with working WebGL2 or WebGPU. All five members completed the mandatory workshops. Sponsor resources and judge access still need confirmation.

178 words / maximum 200

4. Preliminary Environmental & SDG ImpactFramework

The figures below describe a proposed pilot, not results we have already measured. We are assessing career preparation and learning access; we are not claiming reductions in emissions, energy use or national underemployment.

Metric Category	Your Response
Target SDG Sub-Target(s)	Primary 4.4: increase relevant technical and vocational skills for employment, decent jobs and entrepreneurship [R2]. Secondary 8.5: productive employment and decent work [R3]. Career planning and sustained learning are intermediate contributions, not proof of job placement.
Current Problem Baseline	DOSM reported graduate skill-related underemployment of 32.2% in Malaysia in 2024 [R1]. Our pilot baseline has not been measured. First, time how long each of 10 volunteers takes to prepare a valid three-step learning plan manually. The national figure provides context; it is not our pilot baseline or a promised reduction.
Core Measurable KPI	Primary KPI: median minutes to produce a valid three-step plan linked to the same synthetic profile and job advertisement. Percentage improvement = 100 x (baseline median - assisted median) / baseline median. A valid plan names a target, identifies at least two relevant skill gaps, and proposes three actionable learning steps; assess with a fixed rubric.
Phase 2 Impact Target	We propose a 20% reduction in median planning time against the manual baseline, with at least 80% of 10 volunteers completing the core journey without help. Use equivalent profiles and counterbalance task order. Report the baseline, sample size, failures and raw results. These targets still need team approval; they are not achieved or validated outcomes.

Latest-revision note: job-market collection and historical context are implemented. Missing career-history cells may use mock values without a visible label; these cells must not support impact claims. Public deployment of the collector remains unverified.

References

[R1] Department of Statistics Malaysia, Graduates Statistics 2024, underemployment tables/infographic. https://www.dosm.gov.my/uploads/release-content/file_20251031111142.pdf
[R2] United Nations, Goal 4, Target 4.4. https://sdgs.un.org/goals/goal4
[R3] United Nations, Goal 8, Target 8.5. https://sdgs.un.org/goals/goal8

5. Eligibility & Readiness Checklist

Template checklist item	Draft status
All members completed mandatory GenAI, Database and Cloud	YES - completed by all 5 team members;

Deployment workshops.	confirmed by the team.
Project addresses a verified real-world problem mapped to UN SDGs.	Supported by the cited DOSM context and proposed SDG mapping; team to confirm final selection.
Proposed solution targets a functional deployed application, not only a business case.	Yes. The repository contains a runnable game, API and messaging integration. The team supplied the live URL https://xmum-hackfinal.replit.app/; judge access still requires verification.
All fields are complete within word limits.	The answers meet the word limits. Resolve the remaining CONFIRM items before signing.

Before submitting

Team details, Domain 1 and the live URL have been supplied. All five members have completed the mandatory workshops. Confirm the registered SDG track, sponsor provision, rights to bundled assets, and the date on which team-owned work began. Review every declaration personally. Do not sign this draft as a final submission while confirmation items remain.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

CONFIRM Yes or No. We still need each member to identify any work they created or owned before the hackathon. If there is pre-existing work, record its owner, creation date, files and role in the submission. The Bruno Simon game foundation is third-party work, declared separately in Section 6.2. Its engine and original world must not be presented as the team's own creation.

6.2 Third-Party Code, Datasets or Models

Yes. KerjayaKu uses third-party code, assets, services and pretrained AI. The team's additions cover career planning, AI integration, roadmap and progress rules, the multilingual career interface, mentor and expert interactions, pet behaviour and automation. The team still needs to confirm who created these additions and when.

Name	Type	License / terms
Bruno Simon folio-2025 foundation	Code / world assets	Code: MIT, copyright 2025 Bruno Simon; retain GamePlat/game/license.md. Confirm rights for each bundled artwork, audio, font and other asset separately.
Three.js; Vite; GSAP; Howler; Rapier; other dependencies	Code	Package-specific licenses. Full locked JavaScript dependency inventory accompanies the technical declaration; GSAP has its own license terms.
DeepSeek API and configured pretrained model	Model / service	Provider API and model terms; no team training or fine-tuning.
Career/skill reference data and synthetic demo activity	Self-authored reference / synthetic data	Team to confirm authorship. Reference demand/difficulty values are not measured hiring statistics.
Expert perspective cards and public research	Derived text / reference material	Public sources retain their original rights. Nuwa method carries an MIT notice. No endorsement by named individuals.
JSearch responses; Telegram; optional cloud integrations	External data / services	Provider terms and account permissions; only claim live use after confirming deployment and actual calls.

6.3 Employment-Related IP Conflicts

CONFIRM Yes or No after checking with every member. If an employment, internship or consulting agreement could affect ownership, identify the member and agreement, then explain any permission or ownership boundary. This draft does not assume that an employer has given consent.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Leader review required: interpret originality together with the third-party and pre-existing-work disclosures. This draft does not attest to development dates or sign on anyone's behalf.

Field	Response
Signature	
Date	25 October 2026

Appendix A. System Block Diagram

Browser / Three.js game
-> Node.js game API -> direct DeepSeek API
-> local JSON state OR optional authenticated AWS persistence

Telegram -> direct webhook OR n8n forwarding
-> Python FastAPI chatbot -> game integration API
-> DeepSeek classification/chat -> SQLite OR optional AWS storage

Collector -> public/browser sources or JSearch -> sampled ads -> game API
Optional AWS: API Gateway -> Lambda -> Aurora PostgreSQL (Data API)
Secrets Manager holds server credentials; S3 supports migration backups.

Replit deployment configuration is present. Cloud accounts, regions, live calls and public URLs require confirmation. Optional provider code is not evidence of active use.